

Q What is the process that Google follows to identify a project as an open source release?

Most often, the decision to open source a project at Google comes from the product team members themselves. There are a lot of different reasons why they might do this. TensorFlow is a good example of us having worked on machine learning within Google for many years, and then seeing an opportunity to push the entire industry forward by releasing the work that we had done. Because open source has been a part of Google's culture for so long, its benefits are generally well understood throughout the company and so releasing a given project is very natural.

Q Apart from releasing projects for developers, how does Google deploy open source internally to develop innovations?

Internally, most of the source code for all Google products is stored in one monolithic repository. So in many ways, it is like a big open source project where just about everyone can see everyone else's code. This allows Googlers to read others' code to learn from it. It also means that they can submit patches to other teams' projects to help improve them. The new buzzword for this is 'inner source', but for us, it is just how we have always operated.

Beyond that, we also make it really easy for Googlers to bring outside open source code into the company. In terms of comparing just the number of lines of code, we have nearly as much external open source code as we have of code that we've authored ourselves.

Q What are all the leading open source projects that Google leverages to build new solutions?

There is probably not a single Google product that is not touched by open source in some way—whether it is a library used directly by the product, our build and testing tools, or our

data centres that they run in. Big projects like Linux can also be found in our data centres and at the heart of products like Android and Chrome OS. Likewise, LLVM is used extensively as part of the build toolchain, and most of the development languages we use are all open source.

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Q How do open source engineers at Google plan project life cycles?

We have never had a 'one size fits all' approach to open source. Instead, we are comfortable with project heads making the decisions. Most of our projects are developed completely in the open, with very little internal-only discussion or planning. There are certainly cases where development or planning happens internally, and there are different reasons for that. There is a real cost to that style of project management, particularly in terms of community development, so we do our best to help educate teams on the trade-offs and then let them make the decision that is right for their project.

Q What is the average time frame at Google for moving a project from its alpha stage to beta?

Some projects are public right from when the very first line of code is written, while others have been developed internally for quite some time before being released publicly. So projects may be in very different stages of development when they are announced. One thing we are trying to do better is making it more clear what stage a project is in.

Q How do Google engineers test an open source project before its public release?

Some of the bigger releases we have done in the last few years were based on technologies that have existed within Google for many years. For example, TensorFlow was based on our internal machine learning platform called Brain; gRPC is based on our internal RPC system called Stubby; Bazel was called Blaze internally before it was released online; and Kubernetes was the result of our years of experience with our internal job scheduler called Borg. So, in many respects, these projects were tested in the most rigorous manner by first powering Google products and infrastructure.

Now that is not true of all (or maybe even most) of our open source projects. The small- to medium-sized projects go through the same 'dogfooding' trials as Google products, where they are tested internally by fellow Googlers prior to public release.

Q How does Google enable community engagements around its various open source projects?

We generally use the same tools as the rest of the open source community, which includes GitHub, IRC or Slack, Stack Overflow, mailing lists and meetups, among others. Also, we have a strong presence at many open source conferences, and so engage with the community members there or wherever else they happen to be.

We have an extensive network of Google Developer Groups around the world, and these are also a great place for community engagements.

Q How does the Open Source Programs Office develop new policies related to open source projects?

Google's Open Source Programs Office has been around for almost 13 years. So our policies have developed somewhat organically over the years as the company has grown and new